

Computational Methods in Neuroscience - assignment 5

1. B. According to our previous findings of the mutual information, the 5th neuron gives the best indication of the hand's movement direction according to the number of spikes.

$$2. p(x|k) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$X = \{2, 3, 5, 4, 8\}, p_0(\theta) = e^{-k}$$

$$\begin{aligned} \theta_{MAP} &= \operatorname{argmax} p(x|k) = \operatorname{argmax} p(x|\theta) \times p_0(\theta) \\ &\rightarrow \text{Assuming independent distributions: } \theta_{MAP} \\ &= \operatorname{argmax} \left[p_0(\theta) \times \prod_{i=1}^5 p(x_i|\theta) \right] \\ &\rightarrow \text{appling logarithmic scale: } \operatorname{argmax} [\ln(p_0(\theta)) \\ &\quad + \sum_{i=1}^5 \ln(p(x_i|\theta) \times p_0(\theta))] = \{k = \theta\}, \ln(e^{-k}) \\ &\quad + \sum_{i=1}^5 \ln\left(\frac{1}{\theta} e^{-\frac{x_i}{\theta}} \times e^{-\theta}\right) = 5 \ln\left(\frac{1}{\theta}\right) + \sum_{i=1}^5 \left(\frac{x_i + \theta}{\theta}\right) \\ &\quad - (2 + 3 + 5 + 4 + 8) \\ &= 5 \ln\left(\frac{1}{\theta}\right) \\ &\quad + \sum_{i=1}^5 \left(\frac{x_i}{\theta} + 1\right) = 5 \ln\left(\frac{1}{\theta}\right) \\ &\quad + \sum_{i=1}^5 \left(\frac{x_i}{\theta}\right) + 5 = 5 \ln\left(\frac{1}{\theta}\right) + \frac{1}{\theta} \times \sum_{i=1}^5 (x_i) = 5 \ln\left(\frac{1}{\theta}\right) + \frac{22}{\theta} \rightarrow \frac{\partial f}{\partial \theta} \\ &= 5[-1 \times \ln \theta]' + 22\theta^{-1} = \frac{5}{\theta} - \frac{22}{\theta^2} \rightarrow \frac{\partial f}{\partial \theta} = 0 \rightarrow \theta_{MAP} = \frac{22}{5} = 4.4 \end{aligned}$$

b. Yes, it will be affected. In these conditions we'll reach the **expected value** of X.

$$\theta_{MAP} = \frac{55}{6} = 9\frac{1}{2}$$

c. As all the observations have an interval larger than θ can get, we cannot calculate the MAP as it will deal only with observations that their probability is zero.

$$\begin{aligned} d. \quad \theta_{ML} &= \operatorname{argmax} [p(x|\theta)] = \operatorname{argmax} [\prod_{i=1}^5 p(x_i|\theta)] = \operatorname{argmax} 5 \frac{1}{\theta} e^{-\frac{2+3+5+4+8}{\theta}} = \\ &= \operatorname{argmax} 5 \frac{1}{\theta} e^{-\frac{22}{\theta}} \rightarrow \operatorname{argmax} [\ln(5 \frac{1}{\theta} e^{-\frac{22}{\theta}})] = \operatorname{argmax} [(\ln(5 \frac{1}{\theta}) - \frac{22}{\theta} \ln(e))] = \\ &= \operatorname{argmax} [(\ln(5 \frac{1}{\theta}) - \frac{22}{\theta})] \end{aligned}$$

$$\frac{\partial f}{\partial \theta} = -\frac{5}{\theta} + \frac{22}{\theta^2} \rightarrow \frac{\partial f}{\partial \theta} = 0 \rightarrow -\frac{5}{\theta} + \frac{22}{\theta^2} = 0 \rightarrow \theta_{ML} = \frac{22}{5} = 4.4$$

3.

$$X = \{x_1, x_2, \dots, x_n\}$$

$$a. p(x|\theta) = \frac{\theta}{x!} \times e^{-\theta}$$

$$\theta_{ML} = \operatorname{argmax}_{\theta} \sum_{i=1}^n [X_i \ln(\theta) - \theta] = \operatorname{argmax}_{\lambda} [-n \times \theta + \ln(\theta) * \sum_{i=1}^n [X_i]]$$

$$\frac{\partial f}{\partial \theta} = n + \frac{\sum_{i=1}^n X_i}{\theta}, n + \frac{\sum_{i=1}^n X_i}{\theta} = 0 \rightarrow \theta_{ML} = \frac{\sum_{i=1}^n X_i}{n}$$

$$b. p(x|\theta) = \frac{1}{\theta} \times e^{-\frac{x}{\theta}}$$

$$\theta_{ML} = \operatorname{argmax} [p(x|\theta)] = \operatorname{argmax} [\prod_{i=1}^n p(x_i|\theta)] = \operatorname{argmax} \left[\prod_{i=1}^n \left(\frac{1}{\theta} \times e^{-\frac{x_i}{\theta}} \right) \right] \rightarrow$$

$$\operatorname{argmax} \left[\sum_{i=1}^n \ln \left(\frac{1}{\theta} \times e^{-\frac{x_i}{\theta}} \right) \right] = \operatorname{argmax} \left[n \times \left(\ln \left(\frac{1}{\theta} \right) + \ln \left(e^{-\frac{\sum_{i=1}^n x_i}{\theta}} \right) \right) \right] = \operatorname{argmax} \left[n \ln \left(\frac{1}{\theta} \right) - \frac{\sum_{i=1}^n x_i}{\theta} \right]$$

$$\frac{\partial f}{\partial \theta} = -n \frac{1}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} \rightarrow \frac{\partial f}{\partial \theta} = 0 \rightarrow -n \frac{1}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} = 0 \rightarrow \theta_{ML} = \frac{\sum_{i=1}^n x_i}{n}$$

$$c. p(x|\theta) = \theta x^{\theta-1}$$

$$\theta_{ML} = \operatorname{argmax} [p(x|\theta)] = \operatorname{argmax} [\prod_{i=1}^n p(x_i|\theta)] =$$

$$\operatorname{argmax} \left[\prod_{i=1}^n \theta x_i^{\theta-1} \right] \rightarrow \operatorname{argmax} \left[\sum_{i=1}^n \log(\theta x_i^{\theta-1}) \right] = \operatorname{argmax} \left[\sum_{i=1}^n (\log \theta + (\theta - 1) \log x_i) \right]$$

$$\frac{\partial f}{\partial \theta} = n \frac{1}{\theta} + \sum_{i=1}^n \log x_i \rightarrow \frac{\partial f}{\partial \theta} = 0 \rightarrow n \frac{1}{\theta} + \sum_{i=1}^n \log x_i = 0 \rightarrow \theta_{ML} = -\frac{n}{\sum_{i=1}^n \log x_i}$$